

Sales Analysis

Dashboard Report

Project Type: Sales & Business Analytics

Dataset: Superstore Sales Dataset

Tools Used: Python · Pandas · SQL · Seaborn · Power BI

Date: June 2026

Python

Pandas

NumPy

Seaborn

Matplotlib

SQL/SQLite

Power BI

EDA

SECTION

Executive Summary

This report presents a comprehensive sales performance analysis of the Superstore dataset containing **9,994 transaction records** across 21 fields spanning **2014–2017**. The analysis combines Python EDA, SQL querying, and interactive dashboard creation to derive actionable business insights. Total revenue stands at **\$2,297,200.86** with a total profit of **\$286,397.02** across **5,009 unique orders**.

Key findings include the dominance of the West region, Technology as the top profit contributor, and a critical discount-profit inverse relationship that is actively eroding margins.

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SECTION 1

Project Overview & Objectives

This project performs a full-cycle sales data analysis on the Superstore dataset — one of the most widely used retail analytics benchmarks. The analysis spans three layers: Python-based EDA, SQL-based querying with SQLite, and interactive dashboard creation — demonstrating a complete real-world analytics workflow.

- Analyze overall sales performance across regions, categories, and customer segments.
- Identify the most and least profitable products, categories, and sub-categories.
- Examine the impact of discounting on profitability using scatter analysis and SQL aggregation.
- Study monthly and yearly sales trends to identify seasonality and growth patterns.
- Segment customers and identify top-performing customers by total revenue.
- Detect loss-making orders and products — and quantify the magnitude of losses by category.
- Engineer analytical features (Year, Month, Profit Ratio) to enrich time-series analysis.
- Build an interactive sales dashboard with KPI cards, regional maps, and trend charts.
- Translate findings into prioritized, actionable business recommendations.

SECTION 2

Dataset Description

The Superstore Sales dataset is a popular retail transactions dataset originating from Tableau's sample data. It contains order-level transaction data for a US-based retail company spanning four years (2014–2017), covering furniture, office supplies, and technology products.

Metric	Value	Notes
Total Records	9,994 rows	Order-level transactions, 2014–2017
Unique Orders	5,009	Identified by Order ID
Columns (Original)	21 features	Mix of object, int64, float64 dtypes
Engineered Features	+3 columns	Year, Month, Profit Ratio
Total Sales	\$2,297,200.86	Gross revenue across all orders
Total Profit	\$286,397.02	Net profit; ~12.5% overall margin
Loss-Making Orders	1,871 rows	Orders where Profit < 0
Memory Usage	1.6+ MB	Raw dataset before type optimization
Geography	United States	4 regions: West, East, Central, South

Feature Overview

Category	Features
Order Info	Row ID, Order ID, Order Date, Ship Date, Ship Mode
Customer Info	Customer ID, Customer Name, Segment
Geography	Country, City, State, Postal Code, Region
Product Info	Product ID, Category, Sub-Category, Product Name
Financials	Sales, Quantity, Discount, Profit
Engineered	Year, Month, Profit Ratio

SECTION 3

Data Cleaning & Preprocessing

A structured preprocessing pipeline was applied before any analysis. The dataset was clean with no missing values, but required type conversion, duplicate removal, and date parsing.

Step 1 — Null Check

`df.isnull().sum()` confirmed zero null values across all 21 columns. All 9,994 records were non-null, meaning no imputation was required.

```
df.isnull().sum() # All columns = 0
```

Step 2 — Duplicate Removal

Duplicate rows were checked and removed using `drop_duplicates()` to ensure each transaction is represented only once in the analysis.

```
df.drop_duplicates(inplace=True)
```

Step 3 — Date Parsing

The Order Date column was converted from string to `datetime64` format using `pd.to_datetime()`, enabling year/month extraction for time-series analysis.

```
df['Order Date'] = pd.to_datetime(df['Order Date'])
```

Step 4 — Data Type Verification

`df.info()` confirmed: `float64` (3 cols: Sales, Discount, Profit), `int64` (3 cols: Row ID, Quantity, Postal Code), `object` (15 cols: categorical/text fields). No additional type corrections were needed beyond date parsing.

```
df.info() # Confirms dtypes
```

SECTION 4

Feature Engineering

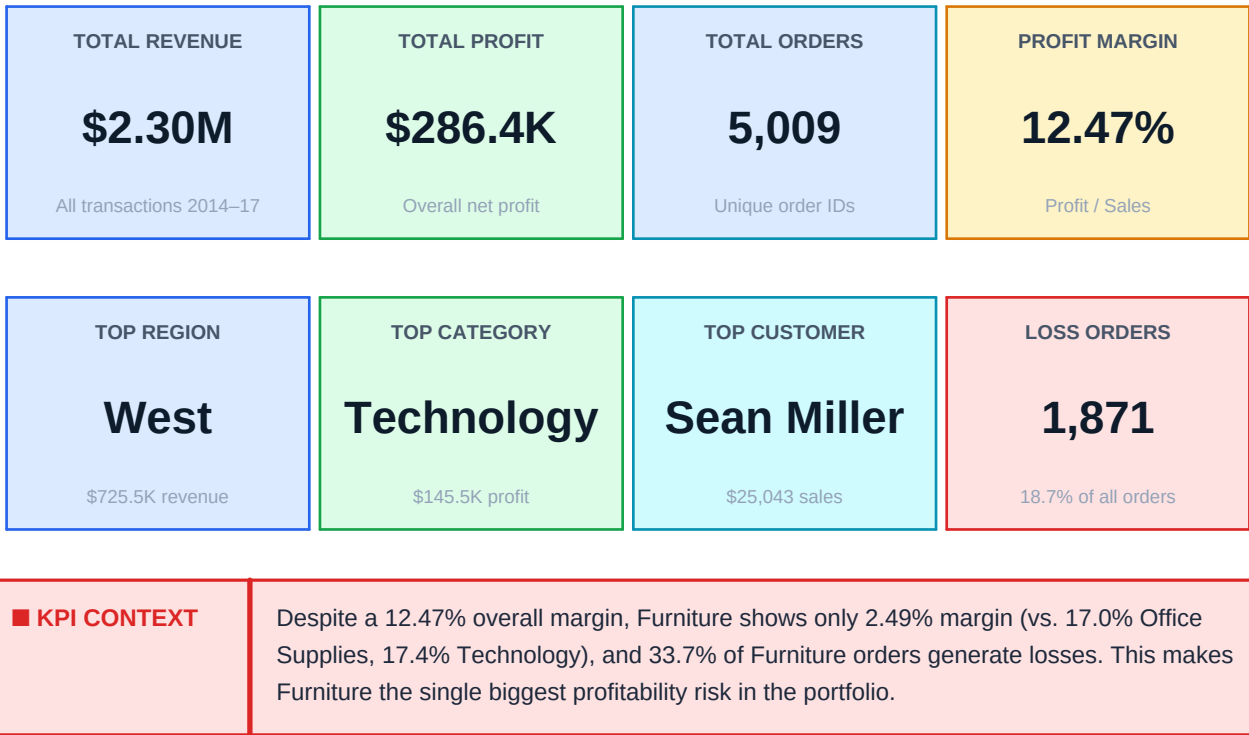
Three analytical features were engineered to enable temporal and profitability analysis beyond the raw transaction fields:

Feature	Formula	Purpose
Year	<code>df['Order Date'].dt.year</code>	Enables year-over-year growth analysis (2014–2017)
Month	<code>df['Order Date'].dt.month</code>	Enables monthly seasonality and trend analysis
Profit Ratio	<code>df['Profit'] / df['Sales']</code>	Margin efficiency metric per transaction (profitability rate)

SECTION 5

Key Performance Indicators (KPIs)

The following headline KPIs were computed directly from the dataset and form the primary metrics for the dashboard:



SECTION 6

Exploratory Data Analysis (EDA)

6.1 Sales by Region

The West region dominates with \$725,457.82 in revenue, followed by East (\$678,781.24), Central (\$501,239.89), and South (\$391,721.91). The South region lags significantly — roughly 46% below the West — suggesting underutilized market potential or a smaller addressable market in that geography.

Region	Total Sales	Share of Total
West	\$725,457.82	31.6%
East	\$678,781.24	29.5%
Central	\$501,239.89	21.8%
South	\$391,721.91	17.1%

6.2 Profit by Category

Technology leads with \$145,454.95 in total profit, closely followed by Office Supplies at \$122,490.80. Furniture is a distant third with only \$18,451.27 — representing just 6.4% of total company profit despite being a major product line. The Technology-to-Furniture profit gap of 7.9x signals a severe profitability imbalance across the product mix.

Category	Total Profit	Profit Margin	Loss Order %
Technology	\$145,454.95	17.40%	14.7%
Office Supplies	\$122,490.80	17.04%	14.7%
Furniture	\$18,451.27	2.49%	33.7%

6.3 Sales by Customer Segment

The Consumer segment is the largest revenue contributor at \$1,161,401.35 (50.6% of total sales), followed by Corporate (\$706,146.37) and Home Office (\$429,653.15). Consumer dominance suggests the company primarily serves individual buyers rather than B2B clients.

Segment	Total Sales	Share
Consumer	\$1,161,401.35	50.6%
Corporate	\$706,146.37	30.7%
Home Office	\$429,653.15	18.7%

6.4 Monthly Sales Trend & Seasonality

Sales exhibit a clear seasonal pattern with strong year-end spikes. September, November, and December consistently show the highest sales volumes across all years. January and February are the weakest months. In 2017, November alone reached \$118,447.83 — nearly 6x January's \$43,971.37. This seasonality should inform inventory planning, staffing, and promotional scheduling.

Period	2014	2015	2016	2017
Jan–Feb (Slow)	~\$18.8K	~\$30.1K	~\$41.5K	~\$64.3K
Sep (Peak 1)	\$81.8K	\$64.6K	\$73.4K	\$87.9K
Nov (Peak 2)	\$78.6K	\$76.0K	\$79.4K	\$118.4K
Dec (Peak 3)	\$69.5K	\$74.9K	\$97.0K	\$83.8K

6.5 Discount vs. Profit Analysis

This is the most critical finding in the analysis. The scatterplot and SQL aggregation reveal a strong inverse relationship between discount level and profitability — orders with discounts of 30% or more consistently generate negative average profit:

Discount Level	Avg. Profit	Status
0% (No Discount)	\$66.90	Profitable
10%	\$96.05	Best Margin
20%	\$24.70	Marginal
30%	-\$45.68	Loss-Making
40%	-\$111.93	Significant Loss
50%	-\$310.70	Critical Loss
80%	-\$101.80	Severe Loss

■ CRITICAL FINDING

Discounts above 20% are destroying profit. The 30%+ discount band accounts for a disproportionate share of the 1,871 loss-making orders. This is likely the #1 fixable driver of the Furniture category's poor margins and overall profit leakage.

6.6 Top 10 Products by Sales & Profit

The Canon imageCLASS 2200 Advanced Copier dominates both sales (\$61,599.82) and profit rankings. Technology hardware (copiers, printers, videoconferencing) fills the top sales slots. However, several high-selling products such as the GBC DocuBind series appear in both the top sales AND top loss-making lists — a critical warning that volume alone does not guarantee profitability.

Rank	Product	Revenue	Note
1	Canon imageCLASS 2200 Advanced Copier	\$61,599.82	Also #1 profit
2	Fellowes PB500 Electric Punch/Binding Machine	\$27,453.38	Volume driver

3	Cisco TelePresence System EX90 Videoconferencing	\$22,638.48	High-value tech
4	HON 5400 Series Task Chairs for Big and Tall	—	Furniture outlier
5	GBC DocuBind TL300 Electric Binding System	—	Loss-maker too

SECTION 7

SQL Analysis

SQL queries were executed via SQLite (ipython-sql) after loading the cleaned DataFrame into a local database. This demonstrates ability to query structured data programmatically alongside Python, which mirrors real-world analytics workflows.

Total Revenue & Profit

```
SELECT SUM(Sales) AS total_sales, SUM(profit) AS total_profit FROM sales;
```

→ Total Sales: \$2,297,200.86 | Total Profit: \$286,397.02

Sales by Region (Ranked)

```
SELECT region, SUM(sales) AS total_sales FROM sales GROUP BY region ORDER BY total_sales DESC;
```

→ West → East → Central → South

Profit by Category

```
SELECT category, SUM(profit) AS total_profit FROM sales GROUP BY category;
```

→ Technology \$145.5K | Office Supplies \$122.5K | Furniture \$18.5K

Sales by Segment

```
SELECT segment, SUM(sales) AS total_sales FROM sales GROUP BY segment;
```

→ Consumer \$1.16M | Corporate \$706K | Home Office \$430K

Regional Category Cross-Tab

```
SELECT Region, Category, SUM(Sales) AS total_sales FROM sales GROUP BY Region, Category;
```

→ Technology leads in East (\$265K) & West (\$252K)

Discount–Profit Relationship

```
SELECT discount, AVG(profit) AS avg_profit FROM sales GROUP BY discount ORDER BY discount;
```

→ Losses begin at 30% discount; worsen to -\$310 at 50%

Top 3 Products by Revenue (Window Function)

```
SELECT * FROM ( SELECT [Product Name], SUM(Sales) AS total_sales, RANK() OVER (ORDER BY SUM(Sales) DESC) AS rnk FROM sales GROUP BY [Product Name] ) WHERE rnk <= 3;
```

→ Canon Copier, Fellowes Machine, Cisco TelePresence

Loss % by Category

```
SELECT Category, SUM(CASE WHEN Profit < 0 THEN 1 ELSE 0 END)*100.0/COUNT(*) AS loss_pct FROM sales GROUP BY Category;
```

→ Furniture 33.7% | Office Supplies 14.7% | Technology 14.7%

SECTION 8

Key Insights & Findings

#	Finding	Business Impact	Priority
1	West region is #1 in revenue at \$725K (31.6% of total sales)	Regional expansion opportunity	High
2	Technology is #1 profit contributor at \$145.5K with 17.4% margin vs Furniture 2.5%	Product mix rebalancing needed	High
3	Discounts >20% generate negative average profit; 50% discount yields -\$310 avg loss	Major margin erosion	Critical
4	Furniture has 33.7% loss-order rate — highest of all categories	Pricing/cost review needed	Critical
5	Consumer segment drives 50.6% of sales but is under-analysed for profitability	Segment-level margin opportunity	Medium
6	Top customer Sean Miller generates \$25,043 — 20x the average customer	VIP retention priority	Medium
7	Year-end seasonality is strong: Q4 (Sep–Dec) drives ~45% of annual sales	Inventory & promo planning	Medium
8	South region significantly underperforms at \$391K — 46% below West	Geographic growth potential	Lower

SECTION 9

Business Recommendations

Based on the analytical findings, the following recommendations are prioritised by potential business impact:

P1	Discount Cap Policy Enforce a maximum 20% discount threshold across all categories. For Furniture, cap at 15% given its already thin margins. Expected impact: Reduce loss orders by ~30–40%
P1	Furniture Pricing Review Conduct a full cost analysis for the Furniture category. Consider retiring or repricing the 10 worst loss-making SKUs. Expected impact: Recover \$38K+ in Furniture losses
P2	Q4 Demand Planning Pre-build inventory for Sep–Dec peak period. Implement promotional campaigns in August to smooth demand and improve margins during peak. Expected impact: Improve margins during peak
P2	South Region Growth Strategy Analyse why South lags 46% behind West. Investigate market size, competition, or sales coverage gaps. Expected impact: Unlock untapped regional revenue
P2	VIP Customer Retention The top 10 customers generate \$155K in combined revenue. Assign dedicated account managers and offer loyalty incentives. Expected impact: Protect \$155K high-LTV revenue
P3	Expand Technology Portfolio Technology generates 17.4% margins with only 14.7% loss-order rate. Increasing Technology's share of the product mix will improve overall profit margin. Expected impact: Improve overall profit margin

SECTION 10

Conclusion & Next Steps

This project demonstrated an end-to-end data analytics workflow — from raw CSV ingestion through Python/Pandas preprocessing, to SQL-based querying, statistical visualisation, and interactive dashboard creation. The analysis identified two critical, immediately actionable issues: (1) the discount policy is actively destroying margins at levels >20%, and (2) the Furniture category is deeply unprofitable with a 33.7% loss-order rate.

■ KEY TAKEAWAY

Addressing just these two issues — discount capping and Furniture SKU rationalisation — could realistically recover \$50,000–\$80,000 in annual profit without requiring any revenue growth, purely through operational discipline.

Recommended Next Steps

1	Build a profit prediction model using Gradient Boosting/XGBoost to score each order's profitability risk at time of placement.
2	Develop a real-time Streamlit or Power BI dashboard with live filters for region, category, date range, and customer segment.
3	Conduct cohort analysis on customer retention — track whether top customers from 2014 are still purchasing in 2017.
4	Implement ABC analysis on the product catalogue to classify items by contribution to revenue and profit.
5	Integrate with CRM data to map individual sales rep performance against regional and category KPIs.

SECTION 11

Technical Stack & Skills Demonstrated

Tool / Skill	Category	Application in This Project
Python 3.x	Language	End-to-end pipeline: loading → cleaning → analysis → export
Pandas	Data Wrangling	DataFrame operations, groupby, sorting, type conversion, feature engineering
NumPy	Numerical	Statistical computation support
Matplotlib	Visualisation	Line plots (monthly trend), chart customisation
Seaborn	Visualisation	Scatter (discount vs profit), bar charts (region, category)
SQLite + ipython-sql	SQL Analytics	In-notebook SQL: SUM, GROUP BY, RANK(), DENSE_RANK(), CASE WHEN
Power BI	Business Intelligence	KPI cards, regional maps, filters, interactive visuals
Jupyter Notebook	Environment	Reproducible, interactive analysis workflow
Git / GitHub	Version Control	Portfolio hosting, README documentation, version tracking

Recommended GitHub Repository Structure

```
superstore-sales-analysis/  
■■■ notebooks/ sales_analysis_dashboard.ipynb  
■■■ data/ superstore_raw.csv · cleaned_superstore.csv · sales.db  
■■■ dashboard/ Sales_Dashboard.pbix (Power BI file)  
■■■ reports/ Sales_Analysis_Dashboard_Report.pdf (this document)  
■■■ images/ charts exported as .png for README  
■■■ README.md business problem · KPIs · key findings · how to run
```

This report was produced as part of a fresher data analytics portfolio project. All analysis was performed independently using publicly available data. The findings and recommendations reflect conclusions drawn from the dataset and are intended to demonstrate Python EDA, SQL querying, dashboard development, and business storytelling skills.